Graft Manipulation

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19.1 Introduction

Graft manipulation is performed to define and to optimize the volume and cellular composition of stem cell sources like apheresis products, bone marrow, or umbilical cord blood.

Basic manipulations comprise centrifugation procedures for depletion of erythrocytes and volume reduction and are required to cryopreserve grafts in the presence of cryoprotectants like DMSO (dimethyl sulfoxide) (Rowley 1992). These are standard procedures for BM and CB, while apheresis products can usually be cryopreserved without further manipulation.

More complex manipulations are used to optimize the cellular composition and to meet requirements of the individual transplant regimen. Selection of $CD34^+$ or $AC133^+$ progenitors from apheresis or BM has been used to produce concentrated stem cell grafts. In recent years, the selective depletion of unwanted cells like $CD3^+$, $TcR\alpha\beta^+$, or $CD45RA^+$ T-cells and others provides a custom-tailored graft. For both, positive selection and depletion, immunomagnetic cell sorting using monoclonal antibodies and paramagnetic microbeads in combination with semi- or fully automated devices has become the standard technique in most laboratories and increasing the activity of graft engineering has been observed in a recent EBMT surveil across Europe (Witte et al. 2023).

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19.2 Graft Manipulation

19.2.1 Physical Manipulations

19.2.1.1 Volume Reduction

Volume reduction might be necessary in small children and is done by a simple centrifugation process and removal of the supernatant.

19.2.1.2 Washing to Reduce Plasma Antibodies or Anticoagulants

Washing might be necessary in case of unwanted isoagglutinins or to lower the heparin concentration and is also done by centrifugation in a bag or dedicated devices and by exchange of plasma with a suitable solution like 0.9% NaCl. Addition of anticoagulant is not necessary as coagulating agents are washed out by the treatment.

19.2.1.3 Depletion of Erythrocytes

Depletion of erythrocytes is necessary in case of blood group incompatibilities and usually confined to bone marrow. Several procedures are

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employed including centrifugation with an apheresis device (Kim-Wanner et al. 2017) or centrifugation in bags or tubes and subsequent harvest of the buffy coat. Especially in small grafts (<125 mL), a separation using density gradient centrifugation (e.g., Ficoll) can be useful with an even stronger depletion of erythrocytes.

19.2.2 Immunomagnetic Procedures

19.2.2.1 CD34 Enrichment

Enrichment of CD34$^+$ stem cells was the first method which provided grafts with a very low number of T-cells and therefore allowed to avoid GvHD highly effective even in haploidentical HCT (Ringhoffer et al. 2004; Handgretinger et al. 2001).

The method has also been successfully used in MSD and MUD HCT to minimize the rate of GvHD (Pasquini et al. 2012; Lang et al. 2003a) and showed a clear advantage regarding combined cGVHD-free and relapse-free survival compared to unmanipulated grafts in myeloid diseases (Tamari et al. 2018).

Moreover, CD34$^+$ selection is used as a graft backbone to which other cell types (unmanipulated DLI, CD45RA depleted DLI, regulatory and conventional T-cells, and others) can be added.

Enrichment can be performed with the Miltenyi Biotec CD34 reagent system which uses a mAb for the CD34 class 2 epitope and therefore has to be detected by an Ab to a different epitope (normally class 3). Stem cells after separation normally show high purity with extremely low amounts of other contaminating cell types. In some cases, various amounts of monocytes are found without detrimental effect on the graft. Due to the small size of the graft, absolute numbers of contaminating T-cells remain low even if a significant percentage might persist. B-cells are passively depleted as well, whereas CD34$^+$CD19$^+$ B-cell precursors are retained: 1–3% in PB, up to 30% in BM preparations.

Recovery of CD34$^+$ cells is in the range of 50–90% (Schumm et al. 1999).

19.2.2.2 CD133 Enrichment

CD133 detects a slightly smaller subpopulation of CD34$^+$ cells and can also be used for enrichment of stem cells with similar results (Koehl et al. 2002; Lang et al. 2004). This method is currently only rarely used.

19.2.2.3 T-Cell Depletion

Immunomagnetic TCD is technically more demanding than CD34$^+$ enrichment as the processed grafts contain a much higher overall number of cells and even extremely low percentages of contaminating T-cells can endanger the success of the manipulation. Moreover, the correct enumeration of T-cells in a depleted graft is challenging and needs special protocols.

CD3 Depletion

Depletion of CD3$^+$ T-cells provides almost untouched grafts with potential antileukemic effectors (e.g., NK cells) enabling fast engraftment and reliable prevention of GvHD. Prospective phase I/II trials showed low TRM rates after haplo-HCT in combination with toxicity- and intensity-reduced conditioning regimens in children and adults with various diagnoses (Federmann et al. 2012; Lang et al. 2014; Flaadt et al. 2023).

The depletion efficacy can be 0.5 log lower than in CD34$^+$ selection. Since in haplo-HCT residual T-cells should not exceed 50×10^3/kg, it might be occasionally necessary to perform a CD34$^+$ selection with parts of the apheresis to remain below the requested thresholds and to guarantee a sufficient number of progenitor cells (Lang et al. 2014; Federmann et al. 2012; Huenecke et al. 2016).

It should be ensured that during the incubation process, all cells come into contact with the CD3 reagent to avoid unstained T-cells which can impair the result of the depletion significantly. This may happen when transferring stained cells into a second bag system leaving unstained cells in the tubing ends and crinkles of the bag behind. Even smallest amounts of 20–50 µL can contain more T-cells than the whole graft should have.

Analysis of CD3 depleted grafts needs special protocols and has to take into account the rare

number of T-cells among the huge overall number of cells. Therefore, a multigating strategy should be implemented and validated, and T-cells should be determined using several parameters. Exclusion of myeloid cells by CD33 could be helpful as well as the use of CD3 in a bright fluorochrome like APC. Gating can be facilitated by using a "spiked" probe with cells of the negative fraction, and a small percentage of cells from the positive fraction was added to set the gate for subsequent analysis of the negative fraction. For statistical reasons, a minimum of 1×10^6 events should be acquired. To prevent takeover of cells from a previous tube, special care should be taken like flushing the cannula with water before the actual acquisition or to clean the cannula on the outside (Schumm et al. 2013).

TcRαβ Depletion

This procedure removes αβ + T lymphocytes via a biotinylated anti-TcRαβ Ab followed by an anti-biotin Ab conjugated to magnetic microbeads while retaining both γδ + T lymphocytes and natural killer cells in the graft.

Depletion with the TcRab reagent has been shown to be associated with a high depletion efficacy (4.7 log), better than after CD3 depletion (4.0 log) and similar to $CD34^+$ enrichment (4.6 log). Moreover, the results differ less than those after CD3 depletion, resulting in $<50 \times 10^3$/kg infused residual TCRαβ+ T-cells, even in small children (Schumm et al. 2013).

Compared to CD34 selected grafts, a faster expansion was seen for $CD3^+$ and for $CD56^+$ in the early phase after haplo-HCT, probably caused by expansion of co-transfused γδ T-cells and NK cells (Lang et al. 2015). Moreover, clinical trials in children and adults demonstrated a very low incidence of acute and chronic GvHD as well as favorable engraftment and TRM rates in malignant and nonmalignant diseases (Locatelli et al. 2017; Kaynar et al. 2017; Bethge et al. 2022). Therefore, this method is currently the most widely used. Moreover, the approach was successfully used to avoid GvHD also in MSD/MUD HCT (Maschan et al. 2016; Sperl et al. 2022).

Detection of TcRαβ+ T-cells should be done with the same precaution used for CD3 depleted cells, with a minimum of 1×10^6 events and several parameters for the identification of the TcRαβ+ cells. Pregating on CD3-PE vs 7-AAD has been shown to be very helpful as well as gating on TcRαβ and TcRγδ cells in the consecutive dot plot (Schumm et al. 2013).

Graft engineering is a time-consuming process, and a recent retrospective analyses showed that 48 hours is an acceptable time frame. Also freezing and thawing of engineered graft are an option in particular in the time of a pandemic (Nijssen et al. 2023).

CD19 Depletion

Depletion of $CD19^+$ B-cells can be done together with CD3 or TcRαβ depletion and prevents effectively the occurrence of EBV-associated PTLD. Although the threshold dose of contaminating B-cells is still not defined, no cases of PTLD were observed in two multicenter trials with 104 children and adults after infusion of median numbers of 28 and 7×10^3 $CD20^+$ cells/kg BW, respectively (Lang et al. 2014; Federmann et al. 2012).

Alternatively, B-cell depletion can be done in vivo by infusion of therapeutic anti-CD20 mAbs (Locatelli et al. 2017).

Detection of $CD19^+$ B-cells needs special attention as the binding of fluorescence-labeled antibody is impaired when cells were preincubated with the CD19 reagent. Therefore, the detection has to be done with an antibody for CD20 which is co-expressed on B cells (Schumm et al. 2006).

Stem Cell Boosts

Poor graft function after HCT is a relevant complication and is defined as at least bilinear severe cytopenia and/or transfusion requirement, which occurs in a situation of full donor chimerism.

Administration of stem cell boosts from the original donor offers a therapeutic option (Remberger et al. 1998).

To reduce the risk of GvHD, ex vivo TCD procedures as mentioned above are recommended (Olsson et al. 2013). Most experience exists with CD34 selected boosts. Response rates of 80% and a low risk of de novo GvHD between 6% and

22% were observed, even in the case of mismatched donors. Moreover, not only hematological recovery but also recovery of lymphocyte subsets could be improved (Askaa et al. 2014; Mainardi et al. 2018). Add back of donor T-cells can be considered to avoid non-engraftment and enhance immune reconstitution (see Sect. 19.2.3).

19.2.3 Add Back of Donor T-Cells

19.2.3.1 Unmanipulated T-Cells

Unmanipulated T-cells can be added to a backbone of $CD34^+$ selected or T-cell depleted grafts to provide T-cell immunity in various situations. For example, engraftment and immune recovery could be significantly improved without increasing the rate of GvHD by adding a defined number of T-cells to positive selected grafts (Seitz et al., 2018). In the post-transplant phase, unmanipulated T-cells can be used to exert alloreactive effects. The tolerable dose of T-cells varies strongly depending on the HLA disparity, the T-cell chimerism in the patient, and the time after transplantation. In MUD HCT or in haploidentical HCT, it can be helpful to cryopreserve a number of vials with a defined number of T-cells (i.e., 100×10^3 $CD3^+$/kg and 25×10^3 $CD3^+$/kg, respectively). Dosage can vary for MUD and family donors between 1 × 10e5 to 1 × 10e6 cells/kg depending on the time from transplant (3–6 months). For haploidentical donors, lower dosages are recommended (1–2, $5 \times 10e^4$/kg).

19.2.3.2 CD45RA Depletion

DLI with $CD45RA^+$-depleted T-cells takes advantage of the $CD45R0^+$ T-cells which obviously exert little graft-versus-host reaction but can provide antileukemic and antiviral activity. Depletion can be done using the same equipment and reagents for depletion. Depletion is highly effective, and contaminating $CD45RA^+$ cells cannot be found at all (Teschner et al. 2014).

19.2.3.3 Regulatory and Conventional T-Cells

Adoptive transfer of regulatory T-cells has been employed by several groups to reduce GvHD incidence. $CD4^+CD25^+$ regulatory T-cells can be selected from a donor leukapheresis product by a 2-step magnetic selection process: a negative selection of $CD19^+$ and $CD8^+$ cells followed by a positive selection of $CD25^+$ cells. In this case, the final product is highly enriched in regulatory T-cells (Di Ianni et al. 2011). Donor regulatory T-cells can also be obtained with FACS sorting and ex vivo expanded through different approaches. Purity of the regulatory T-cell product and the modalities of its infusion vary depending on the different strategies used (Edinger and Hoffmann 2011; Mancusi et al. 2019). Overall, regulatory T-cell adoptive transfer for GvHD prevention showed promising efficacy in key studies (Brunstein et al. 2011; Brunstein et al. 2016; Meyer et al. 2019). Also, $CD4^+CD25^+$ donor regulatory T-cells can be infused 2–3 days before conventional T-cells and at the time of transplant together with $CD34^+$ cells. The adoptive transfer of donor regulatory T-cells allows for the infusion of a large number of donor conventional T-cells with no or limited use of post-transplant pharmacologic immune suppression. This strategy helped to reduce leukemia relapse and chronic GvHD rates with an acceptable incidence of acute GvHD in pilot trials (Martelli et al. 2014; Pierini et al. 2021). Larger studies are needed to validate such promising approach.

19.2.3.4 Virus-Specific T-Cells

Virus-specific T-cells can be enriched from peripheral blood or an unstimulated apheresis of the original (seropositive) stem cell donor or—if not possible—alternatively from a partially matched third-party donor.

Donor-derived-specific T-cells against ADV-, CMV-, or EBV-associated antigens have been already used in many patients suffering from life-threatening infections post-transplant, and clinical or virological response rates between 70% and 86% in clinical trials and in real world data were observed (Icheva et al. 2013; Feucht et al. 2015; Feuchtinger et al. 2010; Heinz et al. 2023). Thus, antigen-specific T-cells have become a widely used therapeutic option for refractory viral diseases. A large phase III trial is currently ongoing.

The most common technique in the field of graft manipulation is the cytokine capture system which employs the secretion of IFNg after stimulation with appropriate Ag or peptide mixtures for immunomagnetic selection of specific T cells. Simultaneous stimulation with several Ag is possible and generates multispecific T-cells.

The selection procedure can be done with a CliniMACS Prodigy® from a maximum of 1×10^9 cells from a nonmobilized or a mobilized apheresis and yields $0.1–2 \times 10^6$ $CD3^+$ $IFNg^+$ target cells.

Accompanying debris and dead cells require an accurate analysis. Moreover, the small amount of target cells limits the sample size available for analysis, and therefore a single platform procedure including cell count and viability in one measurement is recommended. The first step should be done without washing and includes a cell gate to exclude debris. CD45 and 7-AAD can be used for proper determination of cell viability. A second sample can be analyzed after washing for $CD3^+$, $CD4^+$, and $CD8^+$ numbers and the percentage of $IFNg^+$ cells in these subsets. Bystander cells like B cells, monocytes, and granulocytes can be found in low numbers (Feuchtinger et al. 2006).

19.3 Regulatory Requirements

Graft manipulation is regarded as drug manufacturing in most countries and has to follow the requirements of the EU GMP guidelines, the European Pharmacopoeia, and several EU directives. Therefore, clean room areas are required for the manufacturing and a manufacturing license, and a marketing authorization is mandatory for the distribution of the product. A quality assurance system has to be implemented, and specifications have to be in place for both raw material and drug product. In most cases, volume, cell number, cell dose, viability, and composition are minimum parameters. Sterility in the form of microbiological examination of cell-based preparations according to Pharm. Eu. 2.6.27 has to be shown either before release of the product or, in the case of limited stability, after release. A supply chain must be established to ensure that all agents are available in a timely manner. The equipment used must be qualified regularly and comply with the current state of the art in order to meet all regulatory requirements.

Peripheral blood stem cells from both blood and bone marrow for hematopoietic reconstitution are regarded as non-ATMP.

Key Points

- CD34 enrichment yields stem cell preparations with low contaminating T- and B-cells.
- CD3/CD19 depletion preserves large numbers of NK cells in the grafts.
- TcR αβ/CD19 depletion provides large numbers of NK cells and γδ T cells with very low amounts of TcRαβ T-cells.
- DLI with CD45RA-depleted T-cells might reduce the risk of GvHD.
- Donor regulatory and conventional T-cell adoptive immunotherapy with no post-transplant immune suppression provide low incidence of leukemia relapse and chronic GvHD when employed together with a $CD34^+$ selected graft.
- Virus antigen-specific donor- or third-party-derived T-cells can be utilized post-transplant in patients with therapy-refractory viral infections.

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